



COMP9418 – Advanced Topics in Statistical Machine Learning

W1 – Course Intro

Instructor: Edwin V. Bonilla

School of Computer Science and Engineering

July 23rd, 2018

(Last Update: July 23rd, 2018)

Please make sure you read the course outline asap:

www.cse.unsw.edu.au/~cs9418/outline.html

and check the course website regularly

<http://cse.unsw.edu.au/~cs9418>

Admin Requests

Please note that I do not handle admin requests (e.g. waivers, clashes, enrolments, etc) and will not answer emails or forum questions regarding this. Please use these [online forms](#) instead.

Technical Questions

For technical questions regarding the course contents always use the Forums on the course website. I will not reply emails sent to me directly.

Course Details

Course Code	COMP9418
Course Title	Advanced Topics In Statistical Machine Learning
LiC	Edwin Bonilla
Admin	Astrid Dahl
Classes	<i>Lectures:</i> Monday 12:00-15:00, CLB 5 Timetable for all classes
Consult.	Monday 15:00-16:00, K17 Consultation room G01 (ground floor, by appointment with the course admin)
Units of Credit	6
Course Website	http://cse.unsw.edu.au/~cs9418/index.html

What This Course is about

- Statistical machine learning approaches
 - methods for learning and inference in structured probabilistic models
- Healthy balance of theory and practice
- General topics
 - Exact and approximate inference in probabilistic graphical models
 - Learning in structured latent variable models
 - Posterior inference in non-parametric models based on Gaussian processes

What You Need to Know

- COMP9417 (pre-requisite)
 - *No pre-reqs waiver will be granted this year. So, if you haven't done COMP9417, you will not be able to enrol in COMP9418.*
 - Mathematics
 - Solid mathematical background including linear algebra, basic probability theory, multivariate calculus
 - » E.g. Linear Algebra - MATH2501, Several Variable Calculus - MATH2011, Theory of Statistics - MATH2801
- Programming
 - Python
 - » Other high-level programming language experience may help (C++, Matlab, R)
- Software tools
 - Data wrangling
 - » Unix tools (COMP2041), R, Matlab

Learning Outcomes (1)

1. Derive statistical independence assumptions from a given graphical representation of a probabilistic model
2. Understand and implement exact inference methods in graphical models including variable elimination and the junction tree algorithm
3. Derive and implement maximum likelihood learning approaches to latent variable probabilistic models
4. Understand and implement approximate inference algorithms in graphical models including sampling and variational inference
5. Understand and apply basic methods for structured prediction

Learning Outcomes (2)

6. Understand and apply posterior inference and hyperparameter learning in models based on Gaussian process priors

Teaching Strategies and Rationale

Theory, algorithms and empirical analysis

- Lectures (3h): Main concepts and examples
- Tutorials (2h):
 - Theory (1h) and practice (1h)
 - Bring your own device
- Assignments
- Final exam
- Engagement and blended learning
 - All lectures will be on the course website
 - All tutorial and lab materials (questions before, solutions after) will be on the course website
 - We will use [Kahoot](#) as an engagement tool in the lectures
 - Forum for answering questions using [WebCMS3](#) (do not use email)

Student Conduct – Behaviour

“Students have the responsibility to observe standards of equity and respect in dealing with every member of the University community. This applies to **all activities** on UNSW premises and all external activities related to study and research. This includes behaviour in person as well as behaviour on social media, for example Facebook groups set up for the purpose of discussing UNSW courses or course work.”

“**Behaviour that is considered in breach of the Student Code Policy** as discriminatory, sexually inappropriate, bullying, harassing, invading another's privacy or causing any person to fear for their personal safety is serious misconduct and **can lead to severe penalties, including suspension or exclusion from UNSW.**”

Student Conduct – Plagiarism

“Plagiarism is defined as using the words or ideas of others and presenting them as your own. UNSW and CSE treat plagiarism as academic misconduct, which means that **it carries penalties as severe as being excluded from further study at UNSW**. There are several on-line sources to help you understand what plagiarism is and how it is dealt with at UNSW.”

Plagiarism and Academic Integrity

UNSW Plagiarism Procedure

- Unless otherwise stated, every single piece of assessed material should only be your own work
- You are also responsible that your assignment files are not accessible by anyone but you

Assessment

Item	Topics	Due	Marks
Assignment 1	Weeks 1-2	Week 3	10%
Assignment 2	Weeks 3-6	Week 8	20%
Assignment 3	Weeks 1-10	Week 12	20%
Final Exam	All topics	Exam period	50%

- Dates are tentative
- The overall course mark will be the weighted average of the individual components in the table above.

Special Consideration

- Use the formal channel for [Special Consideration](#) through [MyUNSW](#), including documentation on how your work has been affected.
 - Do not email me. I won't reply if you haven't used this
- If your request is reasonable and your work has clearly been impacted, then
 - for an assignment, you **may** be granted an extension
 - for the Final Exam, you **may** be offered a Supplementary Exam
- Note the use of the word "may". None of the above is guaranteed. It depends on you making a convincing case that the circumstances have clearly impacted your ability to work.

More details on the course website

Resources

Main resource:

- [Book] [Machine Learning: A Probabilistic Perspective](#). Kevin P. Murphy. MIT Press. 2012.

Other:

- [Book] [Bayesian Reasoning and Machine Learning](#) . David Barber. Cambridge University Press. 2012. References to online version (2 Feb 2017).
- [Book] [Gaussian Processes for Machine Learning](#) . Carl Edward Rasmussen and Christopher K. I. Williams. The MIT Press. 2006.
- [Book] [Pattern recognition and machine learning](#) . Christopher M. Bishop. Springer, 2006.
- [Book] [Information Theory, Inference, and Learning Algorithms](#) . David J. C. Mackay. Cambridge University Press. 2003.
- [Book] [Probabilistic Graphical Models: Principles and Techniques](#) . Daphne Koller and Nir Friedman. MIT Press. 2009.

Course Schedule

Week	Lecture
1	Intro to probabilistic modelling
2	Exact Inference in graphical models
3	Learning in graphical models
4	Approximate inference: Variational inference I
5	Approximate inference: Variational inference II
6	Approximate inference: Sampling methods
7	Continuous latent variables
8	Markov and Hidden Markov Models
9	Undirected graphical models
10	Gaussian processes (GP) for regression
11	GP classification and approximations
12	Variational learning of GP models
13	-

Subject to minor changes. See more details of tutorials and assessments on the course website.

Feedback

This course will be evaluated at the end of the semester using the myExperience system.

Please provide respectful feedback during the semester so that I can address any problems ASAP.